Year 9 Mathematics

Unit 4: Constructions

4.1 Constructing Shapes

Constructing Shapes

STARTER · 5 MIN

Date: _____ **Title: Constructing Shapes**

LEARNING OBJECTIVES

- Construct a triangle from its three sides
- Decide if a triangle can be drawn, and use scales
- Construct accurate nets of pyramids

STARTER *Answer in your book. No calculators.*

1. How many degrees are in a triangle?
2. What size is each angle of an equilateral triangle?
3. How many faces does a square-based pyramid have?
4. A scale is 1 cm to 10 m. What is 4 cm?

Starter — Answers

STARTER

Tick or correct in green pen

1. **180°**
2. **60°**
3. **5 faces**
4. **40 m**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can construct a triangle from its three sides	☐
SILVER	I can decide if a triangle can be drawn, and use scales	☐
GOLD	I can construct accurate nets of pyramids	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Constructing Shapes

INTRODUCTION · 3 MIN

'Construct' means use only a ruler and compasses. Leave the arcs showing.

Three sides

Draw the longest side, then arc from each end.

Join the ends to where the arcs cross.

Can it be drawn?

The two shorter sides must add to more than the longest.

3 + 4 > 6: yes

Scale drawing

Divide each real length by the scale.

1 cm to 10 m: 40 m → 4 cm

Nets of pyramids

Construct each triangle face on a side of the base.

Arcs from both ends of the side.

Examples 1 and 2 — Triangles from Three Sides

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint longest side first, arc from each end

Example 1

Construct triangle ABC with AB = 8 cm, AC = 5 cm and BC = 7 cm.

Example 2

Construct an equilateral triangle with sides 6 cm. Measure one angle.

Examples 1 and 2 — Corrections

CORRECTIONS

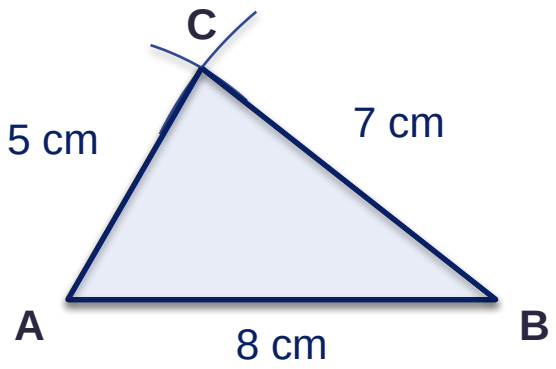
BRONZE

Check your copy and correct in green pen

Example 1

Base first, then two arcs

Arcs: 5 cm from A, 7 cm from B

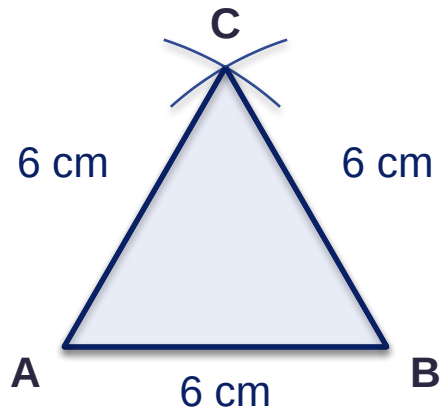


Tip: check: angle A should be 60°.

Example 2

Keep the compasses at 6 cm

Each angle measures 60°



Tip: this is how to construct 60°.

Task 1

PRACTICE · 9 MIN

BRONZE

Construct each triangle. Measure the angle given to check.

Hint longest side first, arc from each end

1. 6 cm, 8 cm, 10 cm
2. 5 cm, 5 cm, 5 cm
3. 4 cm, 7 cm, 9 cm
4. 7 cm, 7 cm, 4 cm
5. 9 cm, 6 cm, 8 cm

★ 4 cm, 7.5 cm, 8.5 cm

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. angle opposite 10 cm = 90°
2. every angle = 60°
3. angle opposite 9 cm ≈ 107°
4. angle opposite 4 cm ≈ 33°
5. angle opposite 9 cm ≈ 79°

★ angle opposite 8.5 cm = 90°

Examples 3 and 4 — Can It Be Drawn? Using Scales

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint shorter sides must add to more than the longest drawing length = real length ÷ scale

Example 3

Can a triangle with sides 8 cm, 15 cm and 7 cm be constructed? Give a reason.

Example 4

A garden is a triangle with sides 30 m, 40 m and 50 m. Construct it using a scale of 1 cm to 10 m.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Add the two shorter sides

8 + 7 = 15

15 is not more than 15

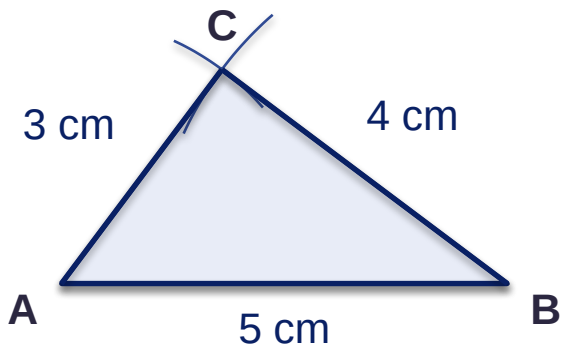
No: the sides lie flat

Tip: the two shorter sides must be longer.

Example 4

Real length ÷ 10

3 cm, 4 cm and 5 cm



Tip: check: the angle opposite 5 cm is 90°.

Task 2

PRACTICE · 9 MIN

SILVER

Questions 1–3: can it be constructed? 4–6: construct using the scale.

1. 3 cm, 4 cm, 8 cm

2. 5 cm, 6 cm, 10 cm

3. 4 cm, 6 cm, 10 cm

4. 20 m, 50 m, 60 m
   scale 1 cm to 10 m

5. 1.5 m, 2 m, 2.5 m
   scale 1 cm to 50 cm

★ 80 km, 60 km, 100 km
   scale 1 cm to 20 km

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. No: 3 + 4 < 8

2. Yes: 5 + 6 > 10

3. No: 4 + 6 = 10

4. 2 cm, 5 cm, 6 cm
(largest angle ≈ 110°)

5. 3 cm, 4 cm, 5 cm
(right angle)

★ 4 cm, 3 cm, 5 cm
(right angle)

Examples 5 and 6 — Nets of Pyramids

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your construction arcs

Hint construct each face with arcs from both ends of a side of the base

Example 5

(6 marks)

Construct the net of a triangular pyramid. The base is an equilateral triangle with sides 4 cm. The other faces have sides 5 cm, 5 cm and 4 cm.

Example 6

(5 marks)

Construct the net of a square-based pyramid. The base has sides 3 cm. The triangles have sides 4 cm, 4 cm and 3 cm.

Examples 5 and 6 — Corrections

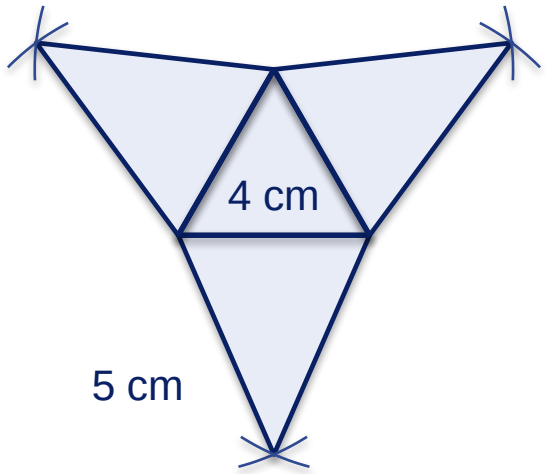
CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

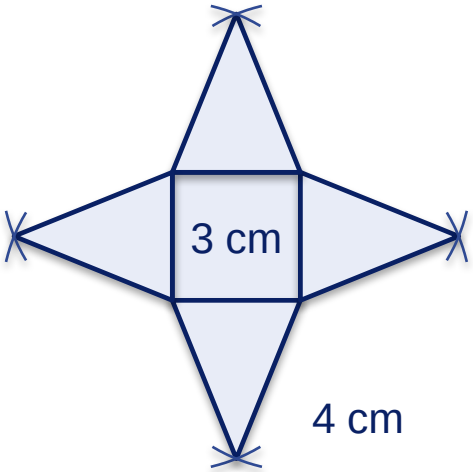
Base first, then a face on each side



Tip: arcs of 5 cm from both ends of each side.

Example 6

Square first, then four triangles



Tip: construct the square's right angles too.

Task 3

PRACTICE · 9 MIN

GOLD

Construct each net. Measure the top angle of one face to check.

Hint construct each face with arcs from both ends of a side of the base

1. Triangular pyramid: all edges 4 cm
2. Triangular: base 3 cm, faces 5, 5, 3 cm
3. Square-based: base 4 cm, faces 5, 5, 4 cm
4. Square-based pyramid: all edges 3 cm
5. Triangular: base 5 cm, faces 6, 6, 5 cm

★ Square-based: base 5 cm, faces 6, 6, 5 cm

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. top angle 60°
2. top angle ≈ 35°
3. top angle ≈ 47°
4. top angle 60°
5. top angle ≈ 49°
★ top angle ≈ 49°

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Construct a triangle with sides 5 cm, 6 cm, 7 cm.

SILVER Can a triangle have sides 2 cm, 3 cm and 6 cm?

GOLD How many triangles in a square-based pyramid's net?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **angle opposite 7 cm ≈ 78°**

SILVER **No: 2 + 3 < 6**

GOLD **4 triangles (and 1 square)**

Exam Practice — Constructing Triangles

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint longest side first, arc from each end

Question 1

(3 marks)

Construct a triangle with sides of length 5 cm, 7 cm and 9 cm.
Show all your construction lines.

Question 2

(2 marks)

Use a ruler and compasses to construct an equilateral triangle with sides of 7 cm.

Exam Practice — Can It Be Drawn?

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint shorter sides must add to more than the longest drawing length = real length ÷ scale

Question 3

(1 mark)

Suraya is asked to construct a triangle with sides 6 cm, 13 cm and 7 cm.
She says this is impossible.
Is Suraya correct? Give a reason.

.....

Question 4

(3 marks)

A field is a triangle with sides 60 m, 80 m and 90 m.
Construct the triangle. Use a scale of 1 cm to 20 m.

Exam Practice — Nets

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint construct each face with arcs from both ends of a side of the base

Question 5

(6 marks)

Construct the net of a triangular pyramid.
The base is an equilateral triangle with sides 3.5 cm.
The other faces have sides 4.5 cm, 4.5 cm and 3.5 cm.

Question 6

(5 marks)

Construct the net of a square-based pyramid. The base has sides 4 cm.
The triangles have sides 4.5 cm, 4.5 cm and 4 cm.

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

one side drawn to ± 2 mm	M1
two arcs crossing, ± 2 mm	M1
triangle completed	A1

Tip: check: the largest angle ≈ 96°

Question 2

7 cm base and two 7 cm arcs	M1
equilateral triangle drawn	A1

Tip: each angle should measure 60°.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

Yes: 6 + 7 = 13, not more B1

Tip: the shorter sides must add to more.

Question 4

3 cm, 4 cm and 4.5 cm B1

one side and arcs drawn M1

triangle completed ± 2 mm A1

Tip: divide each length by 20.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

arcs 3.5 cm from the base ends	M1
equilateral base 3.5 cm	M1 M1
arcs 4.5 cm from one side	M1
one face drawn	M1
other two faces drawn	A1

Tip: top angle of each face ≈ 46°

Question 6

square 4 cm drawn	M1
arcs 4.5 cm from each side	M1 M1
one triangle drawn	M1
all four triangles	A1

Tip: top angle of each face ≈ 53°